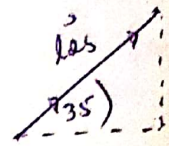
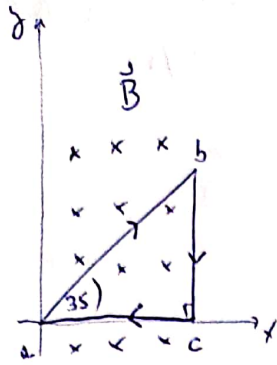


(1)

$$l_{ab} = 0,26 \text{ m}$$

$$\cos 35 = \frac{l_{ca}}{l_{ab}} \Rightarrow l_{ca} = 0,213 \text{ m}$$

$$\sin 35 = \frac{l_{bc}}{l_{ab}} \Rightarrow l_{bc} = 0,149 \text{ m}$$



$$\vec{B} = 0,850 (-\hat{u}) [\text{T}]$$

$$I = 1,15 [\text{A}]$$

$$\vec{l}_{ab} = 0,26 \cos 35 \hat{i} + 0,26 \sin 35 \hat{j}$$

$$\vec{l}_{ab} = 0,213 \hat{i} + 0,149 \hat{j}$$

$$\vec{F}_{ab} = I \vec{l}_{ab} \times \vec{B}$$

$$\vec{l}_{ca} = -0,213 \hat{i}$$

$$\vec{l}_{bc} = -0,149 \hat{j}$$

$$\vec{F}_{ab} = 1,15 \cdot (0,213 \hat{i} + 0,149 \hat{j}) \times (0,850 (-\hat{u}))$$

$$= 0,208 \hat{j} + 0,146 (-\hat{i}) [\text{N}]$$

$$\vec{F}_{bc} = I \vec{l}_{bc} \times \vec{B}$$

$$= 1,15 (-0,149 \hat{j}) \times 0,850 (-\hat{u})$$

$$= 0,146 \hat{i} [\text{N}]$$

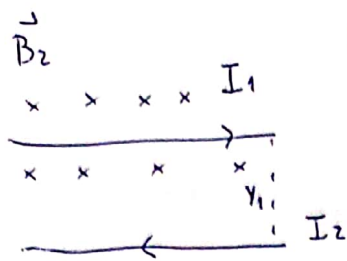
$$\vec{F}_{ca} = I \vec{l}_{ca} \times \vec{B}$$

$$= 1,15 (-0,213 \hat{i}) \times 0,850 (-\hat{u})$$

$$= 0,208 (-\hat{j}) [\text{N}]$$

(2)

$\hat{y}$
 $\hat{z} \rightarrow -\hat{x}$



e)

$$\vec{F}_1 = I_1 \vec{l}_1 \times \vec{B}_2 \quad (3)$$

$$\vec{F}_1 = I_1 l \hat{i} \times \frac{\mu_0 I_2}{2\pi y_1} (-\hat{k})$$

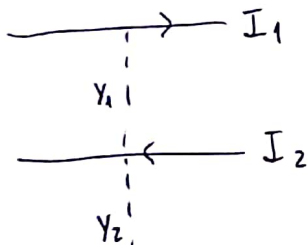
$$\vec{F}_1 = \frac{\mu_0 I_1 I_2 l}{2\pi y_1} \hat{j}$$

$$\Rightarrow \frac{\vec{F}_1}{l} = \frac{4\pi \cdot 10^{-7} \cdot 0,800 \cdot 1,200}{2\pi \cdot 0,08} \hat{j}$$

$$= 2,4 \cdot 10^{-6} \hat{j} \text{ [N/m]} \quad (4)$$

$$b) \vec{F} = q \vec{v} \times \vec{B}$$

$$\vec{v} = 100 \hat{i} \text{ [m/s]}$$



$$\vec{B}_1 = \frac{\mu_0 I_1}{2\pi (y_1 + y_2)} (-\hat{k}) \quad (3)$$

$$\vec{B}_2 = \frac{\mu_0 I_2}{2\pi y_2} \hat{i} \quad (3)$$

$$\Rightarrow \vec{B} = \vec{B}_1 + \vec{B}_2$$

$$= \frac{\mu_0}{2\pi} \left[\frac{I_1}{(y_1 + y_2)} (-\hat{k}) + \frac{I_2}{y_2} \hat{i} \right]$$

$$= \frac{4\pi \cdot 10^{-7}}{2\pi} \left[\frac{0,800}{0,2} (-\hat{k}) + \frac{1,200}{0,12} \hat{i} \right]$$

$$= 2 \cdot 10^{-7} [4 (-\hat{k}) + 10 \hat{i}]$$

(2)

$$= 1,2 \cdot 10^{-6} \hat{k} \text{ [T]}$$

$$\Rightarrow \vec{F} = 48 \cdot 10^{-9} \cdot 100 \hat{i} \times 1,2 \cdot 10^{-6} \hat{k}$$

$$= -5,76 \cdot 10^{-12} \hat{j} \quad [N]. \quad (5)$$

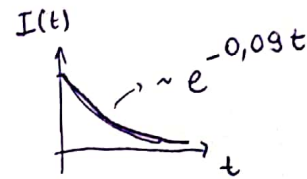
(3.) a) $B = \mu_0 n I$ (2) : Campo del solenoide

$\phi = \int_R \vec{B} \cdot d\vec{S}$: Flujo a través de la bobina. (1 vuelta)

$$= \int_R B dS$$

$$= B \int_R dS$$

$$= B (\pi R^2)$$



$$\Rightarrow \mathcal{E} = -N \frac{d\phi}{dt}$$

$$N = 20.$$

$$= -N \pi R^2 \mu_0 n \frac{dI}{dt} \quad (2)$$

$$= -\mu_0 n N \pi R^2 \frac{d}{dt} (0,2 e^{-0,09t})$$

$$= -4\pi \cdot 10^{-7} \cdot 300 \cdot 20 \cdot \pi \cdot (0,044)^2 \cdot 0,2 \cdot -0,09 e^{-0,09t}$$

$$= 8,254 \cdot 10^{-7} e^{-0,09t} \quad [V]. \quad (3)$$

b)
Dado que I disminuye en "t", el flujo y el campo también disminuyen con "t". Por tanto, la corriente inducida tiene la misma dirección de circulación que el solenoide, de manera de evitar cambios en el flujo. $I = \frac{|\mathcal{E}|}{R} \Rightarrow (5) \quad I = 8,254 \cdot 10^{-8} e^{-0,09t} \quad [A]$